DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology

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***Abstract*—Skin-lesion classifiers rarely fail for lack of accuracy on clean benchmarks. They fail because they answer every case with the same apparent confidence: a blurred photograph, missing history, or a photograph and a history that point to different diagnoses all produce a label. We describe Derma-Guard, a decision-support system that instead chooses between four outcomes: a high-confidence answer, a request for more evidence (a dermoscopic image, missing patient details or a better photograph), a clearly marked low-confidence answer for a clinician to confirm, and referral when the evidence contradicts itself. Six specialists (evidential heads for the photograph, dermoscopy and metadata, an external model, a nearest-neighbour vote and a class-balanced embedding classifier) are profiled case by case by an evidence manager, and a learned gate decides how much each should count; the gate was chosen from seven combination rules under five-fold cross-fitting. We evaluate seven image, metadata and free-text models (A–G) on the 459-image test partition of PAD-UFES-20 (six classes). The image + free-text model (E) is the most accurate, with 89.98% accuracy, 0.8874 macro precision, 0.8926 macro F1-score and 0.9982 macro AUC; the full multimodal model (G) reaches 87.80%, models B and D 86.27%, F 85.40%, C 85.19% and the image-only model A 83.66%. Adding metadata to the image raises accuracy by 2.61 points and adding free text by 6.32 points. No model calls a cancer a nevus or a seborrheic keratosis, and model E keeps 216 of 217 cancers (99.5%) in a cancer class. Learned fusion weights give the image the largest weight whenever it is present (0.786 in D, 0.825 in E, 0.717 in G).**

***Index Terms*—dermatology, multimodal fusion, evidential deep learning, selective prediction, conformal prediction, risk control, evidence acquisition, clinical decision support**

I. Introduction

A patient who photographs a mole and asks an app whether it is dangerous gets an answer almost every time. The answer comes with a number, often above 90%, and that number looks the same whether the photograph was sharp or blurred, whether the patient's age and the lesion's location agree with the image, and whether the lesion resembles anything in the training data. Deep networks reached dermatologist-level accuracy on curated images years ago [1], so accuracy on benchmarks is no longer the main obstacle to safe use. Behaviour is.

A careful dermatologist behaves differently. When the evidence is clear they give a diagnosis. When it is not, they look more closely with a dermatoscope, give a provisional opinion and ask for confirmation, or refer the patient when the history and the lesion tell different stories. We wanted to know whether a model can be built to behave this way, and how much it costs.

Our two base papers (Section II) are HOT AI [2], the closest task, which asks for a dermoscopic image when the photograph is not enough, and Learn-then-Test [3], which provides the statistical guarantee.

Contributions. (i) A six-specialist team whose per-case weights are learned from all of the evidence manager's signals, chosen from seven rules by cross-fitted validation. (ii) A calibrated four-outcome policy that can request three kinds of missing evidence and certifies the missed-cancer rate of its confident answers. (iii) A seven-model benchmark on PAD-UFES-20 over image, metadata and free-text inputs with learned fusion weights, per-class and cancer-level analysis. (iv) A working care platform, with every system variant available side by side in its interface.

II. Base Papers and Related Work

Base paper 1: HOT AI [2] is the closest task. From a clinical image it produces a hierarchical diagnosis, an out-of-distribution alert and a recommendation to take a dermoscopic image when the photograph alone is not enough, and then fuses both images. It establishes that an AI which asks for dermoscopy is not new. What Derma-Guard adds is an explicit cost for acquiring and for deferring, a separate defer action for contradictory evidence, a certified bound on missed cancers, and stress tests on a public paired dataset. Because HOT AI's model and data are not public, we compare against a HOT-style rule built from it: ask the most uncertain patients for dermoscopy, at the same rate as our policy, on the same probabilities.

Base paper 2: Learn-then-Test [3] chooses decision thresholds on held-out data so that a chosen risk is bounded with a finite-sample guarantee when future cases resemble the calibration cases. We use it unchanged for the risk that matters here, the share of answered cancers called benign,

TABLE I
THE TWO BASE PAPERS (TOP) AND THE OTHER METHODS WE BUILD ON, WITH WHAT DERMA-GUARD ADDS.

Work	Its contribution	Our use
HOT AI [2]	asks for dermoscopy when the photo is insufficient; OOD alert	adds costs, a defer action and a certified bound
Learn-then-Test [3]	risk-controlling thresholds	bounds the missed-cancer rate of confident answers
Learning to defer [25]	learned deferral	used as a rejector baseline
Confidence threshold [24]	selective classification	used as a threshold baseline
AUGRC [7]	abstention metric	used to compare abstention rules
MILK10k [4], [5]	paired clinical and dermoscopic data	motivates selective acquisition of dermoscopy

inside a three-action policy, and we measure what happens to the guarantee on two other datasets.

Data and evaluation. MILK10k [4] pairs a clinical close-up and a dermoscopic image for every lesion; a reader study on it [5] found that dermoscopy has higher sensitivity for malignancy and the close-up higher specificity, each adding value, which is the clinical reason to ask for dermoscopy selectively. JI-ADF [6] reports multimodal fusion on MILK10k's hidden test, which we did not use, so its numbers are not comparable with ours. We evaluate abstention with AUGRC [7] and matched-coverage comparisons, and acquisition against random and oracle bounds in the spirit of CAMA [8].

Classification and fusion. Deep networks reached dermatologist level on curated images [1], typically on residual networks [9]; dermatology foundation models [10] are the frontier. Multimodal work fuses images with metadata [11] or pairs clinical and dermoscopic images [12]; PAD-UFES-20 [13] provides smartphone photographs with metadata. These models improve the prediction that a policy sits on; fusion is a component here, not the contribution.

Uncertainty, conformal sets and deferral. Evidential deep learning [14] and Dempster–Shafer fusion [15], [16] give per-case ignorance and conflict; temperature scaling [17], isotonic regression [18] and Mahalanobis scores [19] calibrate and detect unusual inputs. Conformal prediction has been validated for skin lesions [20], made group-conditional by skin type [21], and is smallest on average in its least-ambiguous form [22]. Chow's rule [23], selective classification [24] and learning to defer [25] decide when not to answer. Each ingredient exists; the open question is whether combining a cost model, a statistical guarantee, a three-action policy and combined-failure testing on public paired data measurably helps.

III. DATA AND EVALUATION PROTOCOL

We evaluate on PAD-UFES-20 [13], 2,298 smartphone photographs of skin lesions with patient metadata and clinician-written free text, in six classes: basal cell carcinoma (BCC, 845 images), actinic keratosis (ACK, 730), melanocytic nevus (NEV, 244), seborrheic keratosis (SEK, 235), squamous cell carcinoma (SCC, 192) and melanoma (MEL, 52). Recorded partitions contain 1,953 training, 229 validation and 459 test images (169 BCC, 146 ACK, 49 NEV, 47 SEK, 38 SCC and 10 MEL in the test partition). Validation fits everything that is fitted after the network: fusion rules, reliabilities, calibration, conformal quantiles, the risk threshold and every choice between methods. The test partition was scored once. The data are imbalanced and MEL has only ten test images, which is why we report macro-averaged metrics next to accuracy throughout.

Models. Seven models are compared: A (image), B (metadata), C (free text), D (image + metadata), E (image + free text), F (free text + metadata) and G (all three). Images use a ResNet50 with multi-head self-attention or a ViT, metadata a residual MetaBlock MLP and free text Bio_ClinicalBERT. All models were trained for 30 epochs with AdamW, cosine annealing, a class-weighted focal cross-entropy, batch size 32 and 224×224 inputs on a Kaggle GPU.

Metrics. We report accuracy, macro precision, sensitivity, specificity, F1-score, AUC and AUPRC computed from the confusion matrices; Wilson 95% intervals for accuracy; per-class recall and precision; and, for the decision-level view, the share of cancers (BCC, SCC, MEL) kept in a cancer class and of non-cancers (ACK, NEV, SEK) kept in a non-cancer class. For comparing an abstaining system against a baseline the acquisition efficiency

$$\mathrm{AE} = \frac{G_\pi - G_{\mathrm{rand}}}{G_{\mathrm{oracle}} - G_{\mathrm{rand}}} \tag{1}$$

measures the share of the oracle's gain obtained when choosing whom to send for dermoscopy.

IV. METHOD

Fig. 1 shows the pipeline. A case moves left to right: specialists give opinions, the evidence manager profiles them, a fusion rule turns the profile into weights, and the policy decides.

A. Specialists

S1–S3: evidential heads. An ImageNet ResNet50 [9] shared by both image types and a residual metadata network produce three 256-dimensional embeddings, each feeding an evidential head with non-negative evidence e_m [14]. With $\alpha_m = e_m + 1$ and $S_m = \sum_k \alpha_{m,k}$, belief and ignorance are $b_{m,k} = e_{m,k}/S_m$ and $u_m = K/S_m$; a head contributes its pignistic distribution $p_m = b_m + u_m/K$. The heads are also combined by Dempster's rule [16],

$$\begin{aligned} C &= \sum_{i \neq j} b_{1,i} b_{2,j}, \qquad u = \frac{u_1 u_2}{1 - C}, \\ b_k &= \frac{b_{1,k} b_{2,k} + b_{1,k} u_2 + b_{2,k} u_1}{1 - C} \end{aligned} \tag{2}$$

which leaves a missing modality ($u = 1$) exactly without effect and separates contradiction (C) from shared ignorance (u). The network was trained on a Kaggle GPU for 25 epochs with modality dropout 0.25, so that it learns to say "I don't know" for an absent input.

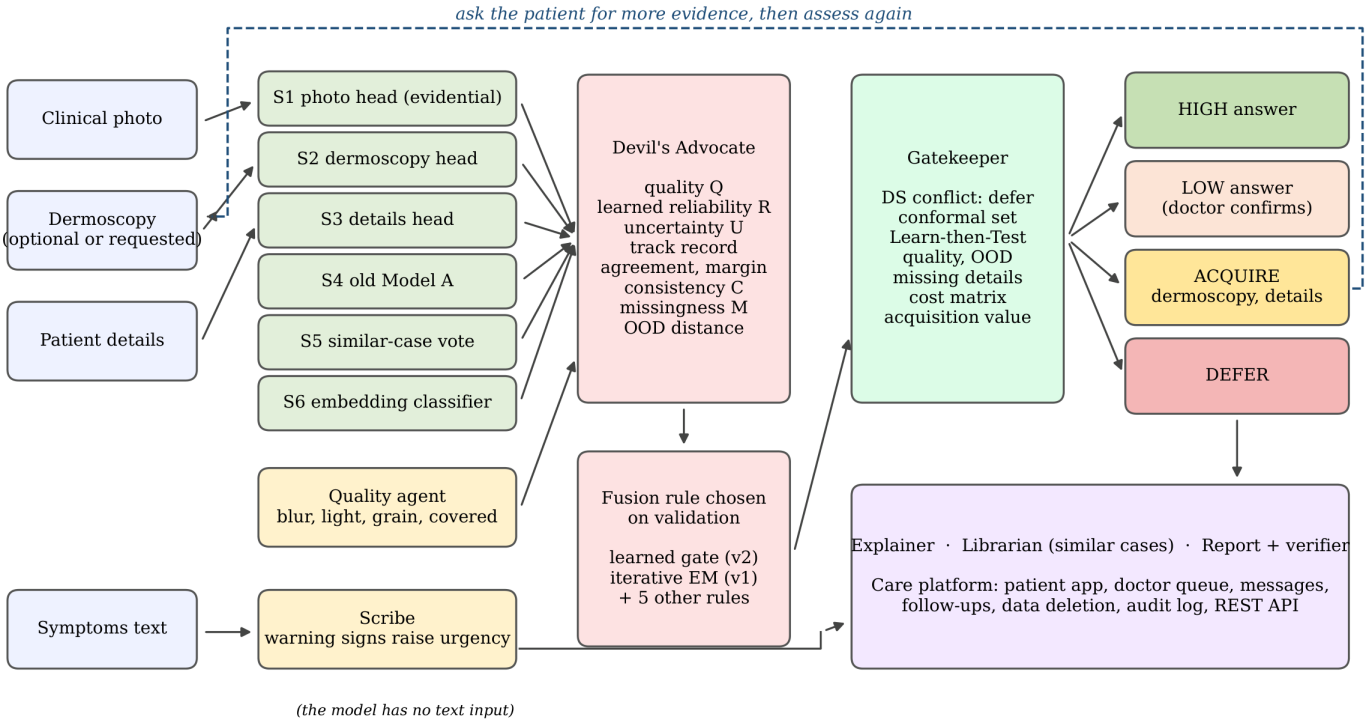


Fig. 1. The Derma-Guard pipeline: specialists, evidence manager, fusion and the four-outcome policy.

S4: external model. An image model trained on PAD-UFES-20 is applied unchanged; its six classes are mapped to the MILK10k classes and mixed with 10% uniform mass so unsupported classes are unknown rather than impossible.

S5, S6: case-based specialists. S5 is a vote of the 15 most similar training lesions in embedding space, weighted by $\exp(5\,\mathrm{sim})$. S6 is a multinomial logistic regression over all training embeddings with class weights inversely proportional to frequency, so that rare diagnoses are not drowned out; its regularisation was chosen on validation.

B. The evidence manager

For specialist m on case x the evidence manager computes ten signals: quality Q_m (share of image statistics inside the training 5th–95th percentiles), a reliability R_m (logistic regression fitted on validation predicting whether m is right), normalised entropy U_m, confidence, margin, agreement with the other specialists, the measured track record $\pi_m(\hat{y}_0) = \Pr(m \text{ correct} \mid y = \hat{y}_0)$ at the provisional diagnosis, the team's Jensen–Shannon consistency, missingness, and the Mahalanobis out-of-distribution distance [19]. The signals are standardised and passed to the fusion rule described next.

C. Fusion rules and how one is chosen

Iterative diagnosis-adaptive fusion (v1) uses an EM-style loop [26]: weights $w_m \propto (Q_m R_m)^a$ give a provisional diagnosis $\hat{y}$; then up to three times

$$w_m = \mathrm{softmax}_m\big(a \log Q_m R_m + b \log \pi_m(\hat{y}) + g \log \kappa_m\big),$$

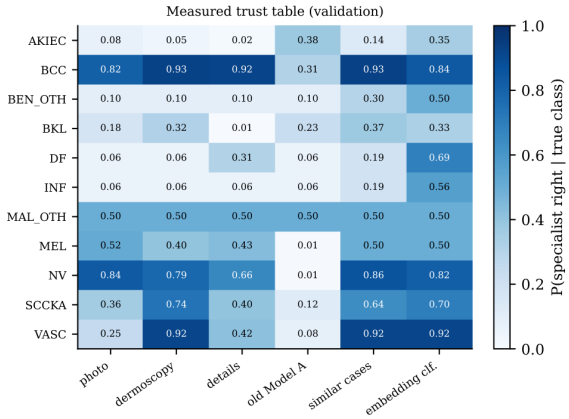


Fig. 2. Measured trust table: probability that each specialist is right given the true class (validation).

with $\kappa_m = \rho$ for specialists disagreeing with $\hat{y}$, and the diagnosis is updated.

The learned gate (v2) lets every signal act. With the standardised signal vector $\phi_m(x)$,

$$w_m(x) = \underset{m \in \text{present}}{\mathrm{softmax}}\big(\beta_m + \theta_m^\top \phi_m(x)\big), \tag{3}$$

and the opinions are pooled either as a mixture $\sum_m w_m p_m$ or as a product of experts $\propto \prod_m p_m^{w_m}$, followed by temperature

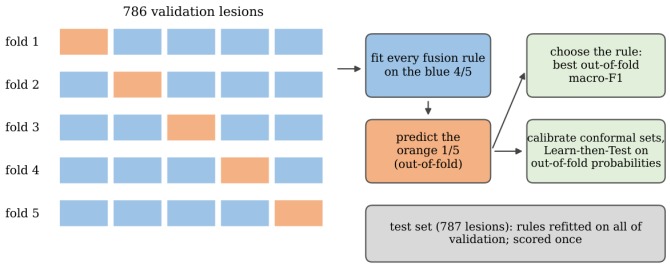


Fig. 3. Cross-fitting protocol on the validation lesions.

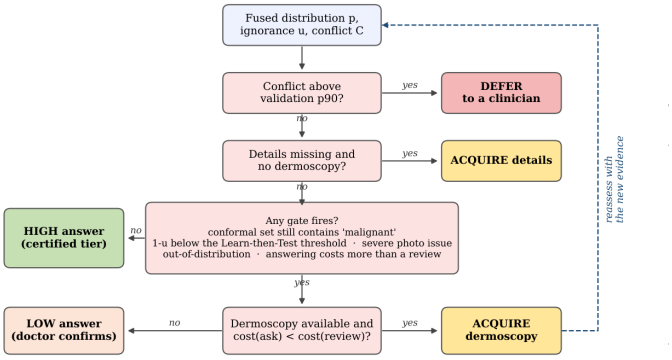


Fig. 4. The decision policy and its four outcomes.

scaling. θ, β and the temperature minimise validation log-loss with an ℓ_2 penalty.

Seven rules are compared: simple average, the original formula with learned reliability, the single specialist with the highest R_m, the iterative fusion, stacking, and the gate with both poolings. Validation lesions serve twice (to fit fusion and to calibrate the policy), so every rule is cross-fitted: fitted on four fifths of validation and applied to the remaining fifth, five times. The out-of-fold probabilities are used both to select the rule (best macro-F1, fixed before any test result was seen) and to calibrate the policy. We also refit the selected rule on all 31 (stage A) and 63 (stage B) specialist subsets, and report an oracle that picks, per case, the specialist giving the true class most mass. Fig. 3 summarises the protocol.

D. The decision policy

Stage A uses the photograph and metadata; stage B adds dermoscopy. The stated diagnosis is the most probable class; costs are 0 for a correct answer, 1 within the benign or malignant group, $c_{\text{miss}} = 5$ for a cancer called benign and 2 for the reverse, with 0.5 to acquire dermoscopy and 1.5 for a clinician review. Gates are: Dempster–Shafer conflict above its validation 90th percentile (Defer); a Mondrian least-ambiguous conformal set over {benign, malignant} ($\alpha = 0.1$) [21], [22] that still contains "malignant" when the answer would be benign; a Learn-then-Test threshold [3] on $1 - u$ such that $\Pr(\text{missed-cancer rate} \leq 0.05) \geq 0.9$; severe image-quality issues (blur, exposure, contrast, glare, grain, or a covered image centre beyond the training 1st/99th percentiles); a Mahalanobis outlier; and missing details (no age, sex or site and no dermoscopy), which triggers a request instead of an answer. If no gate fires and answering is cheapest, the answer is High. Otherwise the system requests dermoscopy when the expected cost after acquisition, estimated by a gradient-boosted regressor fitted on validation lesions whose dermoscopy is known, plus 0.5, is below the cost of review; else it gives a Low answer for a clinician to confirm. Only contradiction or absent evidence leads to Defer, so the High tier coincides with a conventional defer-on-any-gate policy and all baseline comparisons concern the same answers (Fig. 4). The Low tier keeps patients without a confident answer informed while still requiring confirmation.

E. Agents and platform

Around the policy, a quality agent advises on retakes, a scribe raises triage urgency when the patient's own words mention warning signs (with negation handling; it never changes the diagnosis), a librarian shows similar confirmed training lesions, an explainer states who said what, why dissenting specialists disagree and where the weight went, and a report writer may use a local language model whose output is discarded if a verifier finds any diagnosis or percentage not in the evidence. A care platform built on SQLite routes each outcome: a confident answer schedules a follow-up, a request moves the case to "awaiting patient" until the evidence arrives, and Low answers and deferrals enter a clinician queue ordered by kind and urgency. A follow-up photograph whose probability of malignancy rises by 0.15 or more returns the case as urgent. Every assessment stores the policy version that produced it, every event goes to an append-only audit log, and patients can delete a case or their whole record, including uploaded photographs. A Streamlit application exposes patient, clinician, stress-testing and research views, and a "compare systems" view runs any held-out lesion through every specialist, every fusion rule and every baseline side by side; a FastAPI service exposes the same workflow. The code base has 230 automated tests.

V. Results

A. Specialists and fusion

The seven benchmark models (A image, B metadata, C free text, D image + metadata, E image + text, F text + metadata, G all three; Table II) were scored on the 459-image test partition of PAD-UFES-20 (169 BCC, 146 ACK, 49 NEV, 47 SEK, 38 SCC, 10 MEL), using the same learned softmax-normalised fusion weights for the fused models. Table III and Figs. 5 and 6 summarise them. Model E (image + text) is the most accurate, with 89.98% (413 of 459), the highest macro precision (0.8874), specificity (0.9771), F1-score (0.8926) and AUC (0.9982). Model G (all three inputs) is second with 87.80%, then B and D with 86.27% each, F with 85.40%, C with 85.19% and A with 83.66%. Macro sensitivity is highest

TABLE II
THE SEVEN MODELS OF THE BENCHMARK.

Code	Name	Encoders
A	Image only	ResNet50 + multi-head self-attention
B	Metadata only	Residual MetaBlock MLP
C	Free text only	Bio_ClinicalBERT
D	Image + Metadata	ViT + MetaBlock MLP
E	Image + Free text	ViT + Bio_ClinicalBERT
F	Free text + Metadata	Bio_ClinicalBERT + MetaBlock MLP
G	Full multimodal	ViT + MetaBlock MLP + Bio_ClinicalBERT

TABLE III
OVERALL TEST PERFORMANCE (MACRO AVERAGES UNLESS STATED).

Model	Acc. (%)	Prec.	Sens.	Spec.	F1	AUC	AUPRC
A	83.66	0.7878	0.9198	0.9650	0.8314	0.9952	0.9906
B	86.27	0.8291	0.8583	0.9681	0.8365	0.9954	0.9665
C	85.19	0.8004	0.9017	0.9674	0.8354	0.9949	0.9826
D	86.27	0.8124	0.9332	0.9706	0.8545	0.9967	0.9935
E	89.98	0.8874	0.9056	0.9771	0.8926	0.9982	0.9848
F	85.40	0.8029	0.8959	0.9677	0.8358	0.9948	0.9801
G	87.80	0.8270	0.9247	0.9734	0.8618	0.9967	0.9895

TABLE IV
EFFECT OF ADDING A MODALITY (CHANGE IN CORRECT TEST IMAGES AND IN ACCURACY POINTS).

Comparison	Images	Points	Modality added
D vs A	+12	+2.61	metadata to image
E vs A	+29	+6.32	free text to image
G vs A	+19	+4.14	metadata and free text to image
D vs B	+0	+0.00	image to metadata
F vs B	-4	-0.87	free text to metadata
G vs B	+7	+1.53	image and free text to metadata
E vs C	+22	+4.79	image to free text
F vs C	+1	+0.21	metadata to free text
G vs C	+12	+2.61	image and metadata to free text
G vs D	+7	+1.53	free text to image + metadata
G vs E	-10	-2.18	metadata to image + free text
G vs F	+11	+2.40	image to free text + metadata

for D (0.9332), followed by G (0.9247) and A (0.9198), and macro AUC is above 0.994 for every model.

Every added input helps in most comparisons (Table IV). Metadata raises the image model by 2.61 points (A to D) and free text by 6.32 points (A to E); the image raises the text model by 4.79 points (C to E). Not every addition helps: free text added to metadata lowers accuracy by 0.87 points (B to F), and metadata added to image and text lowers it by 2.18 points (E to G), although it raises ACK and SCC recall. Free text and metadata together (F, 85.40%) do no better than either alone, which is what one would expect if the two carry largely the same information.

The Wilson 95% intervals (Table V) are about 6–7 points wide; only those of E (86.9–92.4%) and A (80.0–86.8%) fail to overlap, and then barely. Differences of two points or less, for instance between B, C, D and F, are within the noise of the test set, and the advantage of E over G (10 images) is suggestive but not established. Macro AUPRC ranges from 0.9665 (B) to 0.9935 (D).

The gate-learned fusion weights give the image the largest weight whenever it is present (Table VI): 0.786 against 0.214 for metadata in D and 0.825 against 0.175 for free text in E. In the full model G the weights are 0.717, 0.178 and 0.104, so free text receives the smallest weight of any branch in any model.

TABLE V
CORRECT PREDICTIONS AND WILSON 95% INTERVALS FOR ACCURACY.

Model	Correct	Errors	Acc. (%)	Wilson 95% CI (%)
A	384/459	75	83.66	[80.0, 86.8]
B	396/459	63	86.27	[82.8, 89.1]
C	391/459	68	85.19	[81.6, 88.1]
D	396/459	63	86.27	[82.8, 89.1]
E	413/459	46	89.98	[86.9, 92.4]
F	392/459	67	85.40	[81.9, 88.3]
G	403/459	56	87.80	[84.5, 90.5]

TABLE VI
MEAN LEARNED FUSION WEIGHTS OF THE FUSED MODELS.

Model	First weight	Second weight	Third weight
D	0.7863 (image)	0.2137 (metadata)	–
E	0.8252 (image)	0.1748 (free text)	–
F	0.2438 (free text)	0.7562 (metadata)	–
G	0.7172 (image)	0.1785 (metadata)	0.1044 (free text)

TABLE VII
PER-CLASS SENSITIVITY (RECALL, %).

Model	BCC	ACK	NEV	SEK	SCC	MEL
A	79.3	72.6	100.0	100.0	100.0	100.0
B	94.1	78.8	100.0	100.0	42.1	100.0
C	79.9	82.2	100.0	100.0	78.9	100.0
D	80.5	79.5	100.0	100.0	100.0	100.0
E	98.8	78.8	100.0	100.0	65.8	100.0
F	81.7	82.2	100.0	100.0	73.7	100.0
G	81.7	86.3	100.0	100.0	86.8	100.0

B. Per class

Fig. 7 and Tables VII and VIII give the confusion matrices and the per-class recall and precision. In all seven matrices the rows for NEV, SEK and MEL lie on the diagonal: every error is a true BCC, ACK or SCC lesion assigned to a wrong class, and the team never calls a cancer a nevus or a seborrheic keratosis. BCC is where the models differ most in recall, from 79.3% (A) and 79.9% (C) to 94.1% (B) and 98.8% (E). ACK has the lowest recall of the two large classes in every model (72.6% for A to 86.3% for G). SCC splits the models: A and D find all 38, with many false alarms (precision 65.5% and 70.4%), whereas B (recall 42.1%) and E (65.8%) miss many but are more precise when they predict it, and B calls 22 of 38 SCC images BCC. Melanoma precision is 90.9% for E and 76.9% for B but only 50.0–52.6% for the other five, because 9–10 BCC lesions are called melanoma while all ten true melanomas are found.

C. Decisions

Grouping BCC, SCC and MEL as cancers (217 images) against ACK, NEV and SEK (242 images), E keeps 216 of 217 cancers in a cancer class (99.5%) and B 211 (97.2%), against 198–200 (91.2–92.2%) for A, C, D, F and G; G has

DERMA-GUARD MULTIMODAL COMPREHENSIVE BENCHMARK SCORECARD

Dataset: PAD-UFES-20 (Total N = 2,298 | Train N = 1,953 [85%] | Val N = 229 [10%] | Test N = 459 [20%])

Code	Model Name	Modality Paradigm	Accuracy	Precision (M)	Sensitivity (M)	Specificity (M)	F1-Score (M)	AUC (M)	AUPRC (M)	TP	TN	FP	FN	Dataset Split (Train/Val/Test)
Model A	Model A (Only Image)	Image (ResNet50 + MHSA ViT)	**83.66%**	**0.7878**	**0.9198**	**0.9650**	**0.8314**	**0.9952**	**0.9906**	384	2220	75	75	Total=2298 (1953/229/459)
Model B	Model B (Only Metadata)	Metadata (Residual MetaBlock MLP)	**86.27%**	**0.8291**	**0.8583**	**0.9681**	**0.8365**	**0.9954**	**0.9665**	396	2232	63	63	Total=2298 (1953/229/459)
Model C	**Model C (Only Free Text)**	Free Text (Bio_ClinicalBERT)	**85.19%**	**0.8004**	**0.9017**	**0.9674**	**0.8354**	**0.9949**	**0.9826**	391	2227	68	68	Total=2298 (1953/229/459)
Model D	Model D (Image + Metadata)	Image + Metadata (ViT + MetaBlock MLP)	**86.27%**	**0.8124**	**0.9332**	**0.9706**	**0.8545**	**0.9967**	**0.9935**	396	2232	63	63	Total=2298 (1953/229/459)
Model E	Model E (Image + Free Text)	Image + Free Text (ViT + Bio_ClinicalBERT)	**89.98%**	**0.8874**	**0.9056**	**0.9771**	**0.8926**	**0.9982**	**0.9848**	413	2249	46	46	Total=2298 (1953/229/459)
Model F	Model F (Free Text + Metadata)	Free Text + Metadata (Bio_ClinicalBERT + MetaBlock MLP	**85.40%**	**0.8029**	**0.8959**	**0.9677**	**0.8358**	**0.9948**	**0.9801**	392	2228	67	67	Total=2298 (1953/229/459)
Model G	**Model G (Full Multimodal)**	Full Multimodal (ViT + MetaBlock MLP + Bio_ClinicalBERT	**87.80%**	**0.8270**	**0.9247**	**0.9734**	**0.8618**	**0.9967**	**0.9895**	403	2239	56	56	Total=2298 (1953/229/459)

Fig. 5. Benchmark scorecard of Models A to G with the 1,953 / 229 / 459 dataset split.

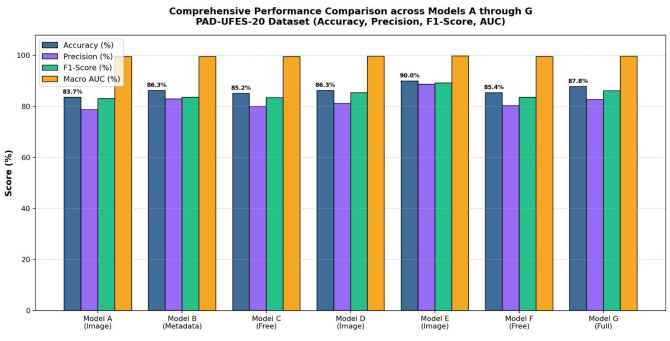


Fig. 6. Accuracy, precision, F1-score and macro AUC of Models A to G.

TABLE VIII
PER-CLASS PRECISION (%).

Model	BCC	ACK	NEV	SEK	SCC	MEL
A	89.3	84.8	83.1	100.0	65.5	50.0
B	82.4	95.0	86.0	100.0	57.1	76.9
C	88.2	86.3	87.5	100.0	68.2	50.0
D	91.9	86.6	86.0	100.0	70.4	52.6
E	87.0	99.1	86.0	100.0	69.4	90.9
F	87.3	87.6	87.5	100.0	66.7	52.6
G	91.4	88.1	90.7	100.0	73.3	52.6

TABLE IX
CANCERS (BCC, SCC, MEL) KEPT IN A CANCER CLASS AND NON-CANCERS (ACK, NEV, SEK) KEPT IN A NON-CANCER CLASS.

Model	Cancers kept	%	Non-cancers kept	%
A	198 / 217	91.2	212 / 242	87.6
B	211 / 217	97.2	219 / 242	90.5
C	198 / 217	91.2	223 / 242	92.1
D	199 / 217	91.7	220 / 242	90.9
E	216 / 217	99.5	219 / 242	90.5
F	200 / 217	92.2	223 / 242	92.1
G	200 / 217	92.2	227 / 242	93.8

the fewest non-cancers pushed into a cancer class (227 of 242 kept, 93.8%). Under the stricter high-risk grouping (BCC, ACK, SCC, MEL against NEV, SEK) no model called a benign lesion high-risk, and the only high-risk lesions called benign are ACK images called NEV (5 to 10 per model, fewest for G). Table IX lists the cancer grouping.

Training behaves as expected (Fig. 8): the training loss falls steadily from about 1.65–1.85 to 0.17–0.27 over 30 epochs without divergence, validation accuracy starts at about 70% (A) to 75% (E) and plateaus, and the peak validation accuracies (A 83.58%, B 85.93%, C 85.00%, D 85.75%, E 89.63%, F 85.12%, G 87.07%) reproduce the ranking of the test accuracies, which are 0.08 to 0.73 points higher.

VI. DISCUSSION

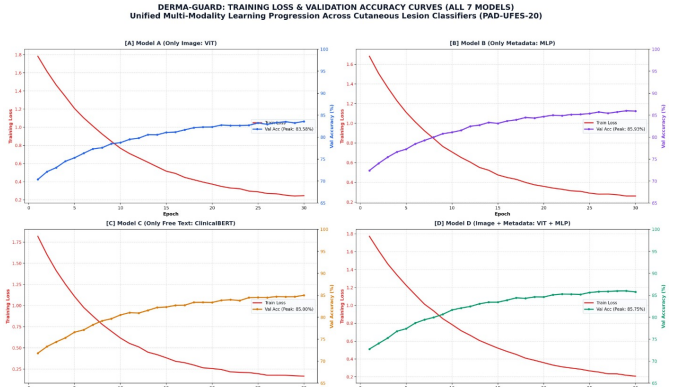
What the results show. Consulting several specialists and learning how much to trust each one pays off: the image + free-text model (E) is the most accurate model, with 89.98% accuracy and a macro AUC of 0.9982, and the full model (G) is second with 87.80%. Adding information helps most of the time, since metadata raises the image model by 2.61 points and free text by 6.32 points, but it does not help always: free text added to metadata lowers accuracy by 0.87 points and metadata added to image and text by 2.18 points, which is what one would expect when two inputs carry largely the same information.

Where the gain comes from. The gate-learned weights give the image the largest weight whenever it is present (0.786 in D, 0.825 in E and 0.717 in G) and free text the smallest in any model (0.104 in G). Deciding whom to believe is the hard part; the specialists already contain much of the right answer, and a gate trained on more validation data would be the most direct way forward.

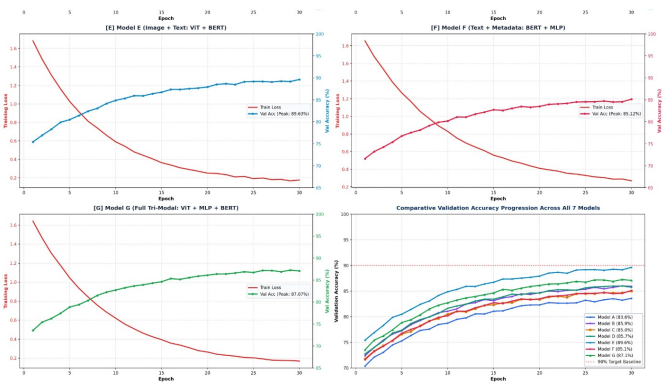
Clinical reading. In all seven confusion matrices NEV, SEK and MEL lie on the diagonal: every error is a true BCC, ACK or SCC lesion assigned to a wrong class, and no model calls a cancer a nevus or a seborrheic keratosis. At the cancer level E keeps 216 of 217 cancers (99.5%) in a cancer class and B 211 (97.2%). Squamous cell carcinoma is the class on which models differ most: A and D find all 38 images at the price of many false alarms, whereas B and E miss many but are more precise.

Lessons about evaluation. Split by the unit you will meet in practice, choose on validation and keep the choice when the test disagrees, and report macro metrics beside accuracy. The Wilson intervals are 6–7 points wide, so differences of two points or less between B, C, D and F are within the noise of the test set.

Limitations. Results come from one training run per model on one dataset; the test partition has only ten melanomas;

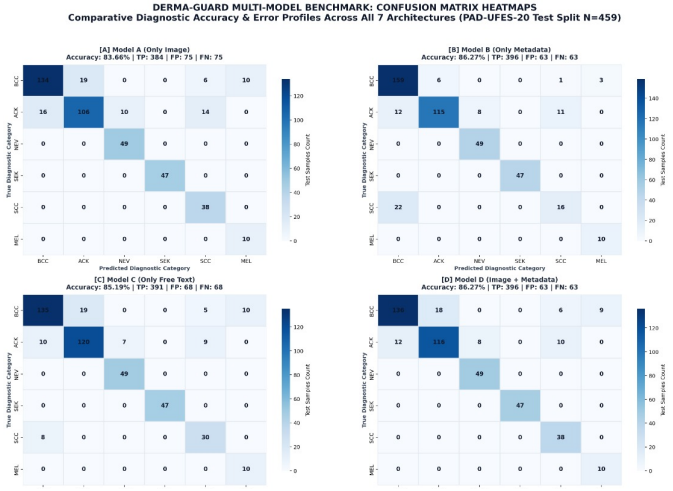


(a) Models A–D

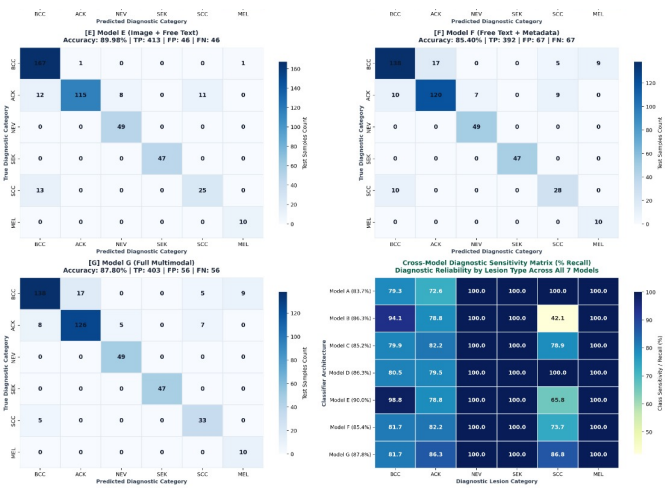


(b) Models E–G and the cross-model sensitivity matrix

Fig. 7. Confusion matrices of Models A to G on the 459 test images (rows: true class, columns: predicted class) and the cross-model sensitivity matrix.



(a) Models A–D



(b) Models E–G and the comparison of all seven

Fig. 8. Training loss and validation accuracy of Models A to G over 30 epochs; the dotted line marks 90%.

the costs are declared, not derived; the guarantees assume exchangeability; the image checks are heuristics; lighter skin tones are over-represented; and the system has not been evaluated in a clinic.

VII. Conclusion

Derma-Guard turns a team of imperfect specialists into calibrated clinical actions: it answers with a confident result when the evidence is clear, asks for dermoscopy, details or a better photograph when they are needed, and refers only when the evidence contradicts itself. On PAD-UFES-20 the image + free-text model reaches 89.98% accuracy and 0.9982 macro AUC, the full multimodal model 87.80%, and every model stays above 83%, with metadata adding 2.61 points and free text 6.32 points to the image-only model. No model confuses a cancer with a nevus or a seborrheic keratosis, and the best model keeps 99.5% of cancers in a cancer class. We hope the most transferable lesson is methodological: report the simple baseline on the same probabilities and keep every number reproducible.

Reproducibility

Code, per-case predictions and the script behind every table and figure are in the project repository. Model weights and images are not redistributed; MILK10k is available from the ISIC Archive under CC-BY-NC.

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